MethReg: estimating the regulatory potential of DNA methylation in gene transcription

Tiago Chedraoui Silva[1,4], Juan I. Young[2,3], Eden R. Martin[2,3], Xi Chen[1,4], Lily Wang[1,2,3,4]

[1]Division of Biostatistics, University of Miami, [2] Dr. John T Macdonald Foundation Department of Human Genetics, [3]John P. Hussman Institute for Human Genomics, [4]Sylvester Comprehensive Cancer Center, University of Miami

UNIVERSITY OF MIAMI

SYLVESTER COMPREHENSIVE CANCER CENTER
UNIVERSITY OF MIAMI HEALTH SYSTEM

BACKGROUND

Epigenome-wide association studies (EWAS) often detect a large number of differentially methylated CpGs, many are located far from genes, complicating the interpretation of their functionalities. Therefore, there is a critical need to better understand the functional impact of these CpGs.

We performed a MethReg analysis for the ROSMAP Alzheimer's Disease (AD) brain dataset with DNA methylation and RNA-seq data from 529 independent subjects. MethReg identified 60 methylation-sensitive transcription factors, many of which are well-known regulators for AD such as TCF12, SPI1, NR3C1, CEBPB, GABPA, and others.

Recent studies demonstrated methylated CpGs can affect gene transcription by either increase or decrease transcription factor binding strengths.

METHOD

We developed a new method and software, MethReg, that analyzes matched DNA-methylation and gene-expression data, along with external transcription factor (TF) binding information, to evaluate, prioritize, and annotate CpG sites with high regulatory potential.

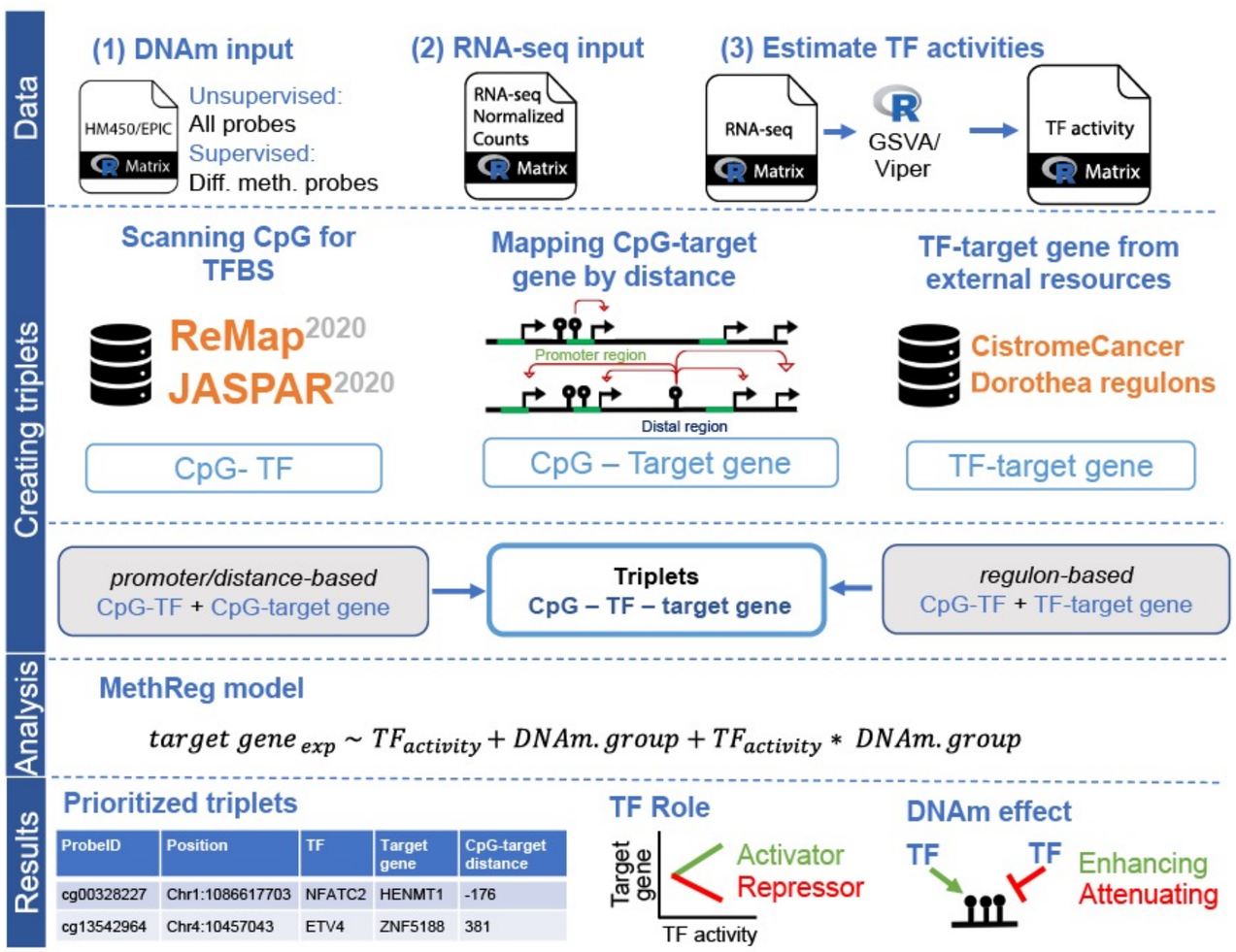
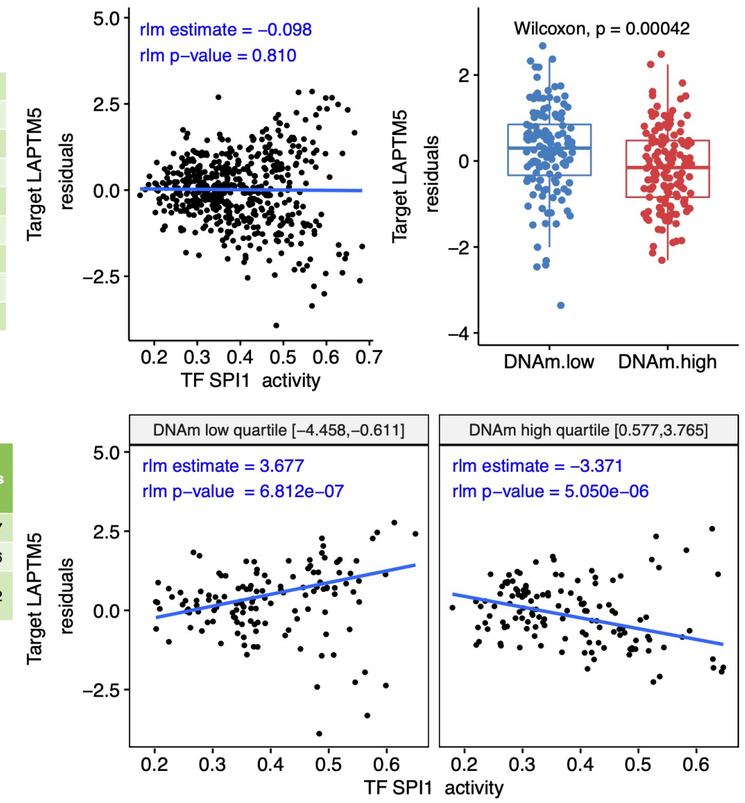


Figure 1 MethReg analysis workflow.

RESULTS

We performed a MethReg analysis for the ROSMAP Alzheimer's Disease (AD) brain dataset with DNA methylation and RNA-seq data from 529 independent subjects. MethReg identified 60 methylation-sensitive transcription factors, many of which are well-known regulators for AD such as TCF12, SPI1, NR3C1, CEBPB, GABPA, and others.

Genome of reference	hg19
Region ID	chr1:31229122–31229123
Probe ID	cg17418085
Target gene ID	ENSG00000162511
Target gene Symbol	LAPTM5
TF gene ID	ENSG00000066336
TF gene Symbol	SPI1
TF role	Dual
DNAm effect	Invert



Target ~ TF + DNAm Quant. Group + TF * DNAm Quant. Group	Estimate	P–Values
Direct effect of DNAm	2.12	2.91e−07
Direct effect of TF	3.6	1.03e−06
Synergistic effect of DNAm and TF	−7.18	7.36e−12

Legend
rlm: robust linear model

Figure 2 An example triplet in which CpG methylation is predicted to invert TF activity on the target gene. In samples with low DNA methylation, target gene expression increases as TF activity increased. In contrast, in samples with high DNA methylation, target gene expression decreased as TF activity increased. Therefore, the TF SPI1 is predicted to have dual modes of regulation depending on CpG methylation levels.

REFERENCES

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Silva, T. C., et al. (2021). "MethReg: estimating the regulatory potential of DNA methylation in gene transcription." https://doi.org/10.1101/2021.02.18.431696

SUPPORT

NIH R01CA158472 R01AG062634 R01CA200987 R21AG060459 R01AG060472 R01AG061127